Flexible Docking and Lead Optimization using Scipion-chem

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1) Flexible docking

Depending on the **flexibility** of the **receptor** and the **ligand**:

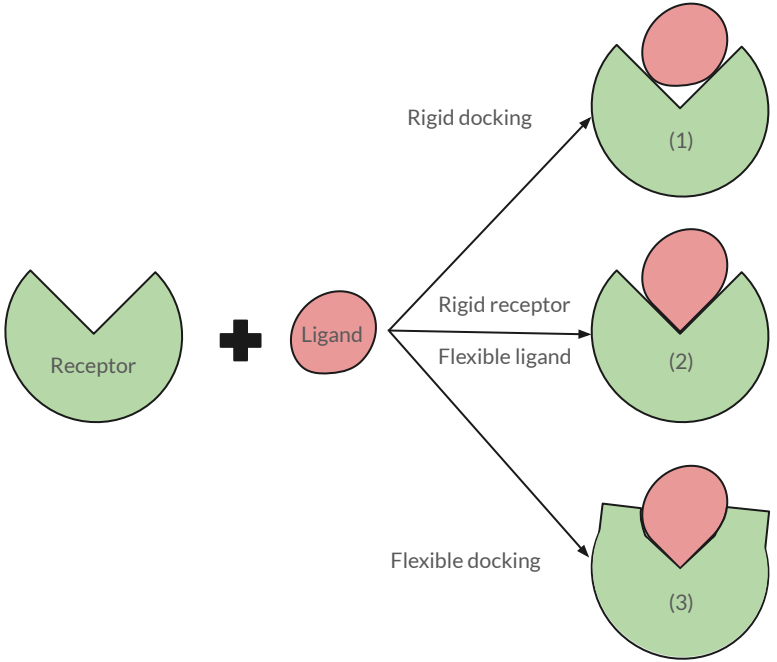
- Rigid (1) // Semi-flexible (2) // Flexible docking (3)

4.1) Flexible receptor residues

- **AutoDock** and **GNINA** can handle **flexible receptor residues** to better fit the ligands during the docking

4.2) Induced Fit Docking

- **Schrodinger** protocol, that combines **soft docking** with **structure minimizations**.
- Several combinations of steps.



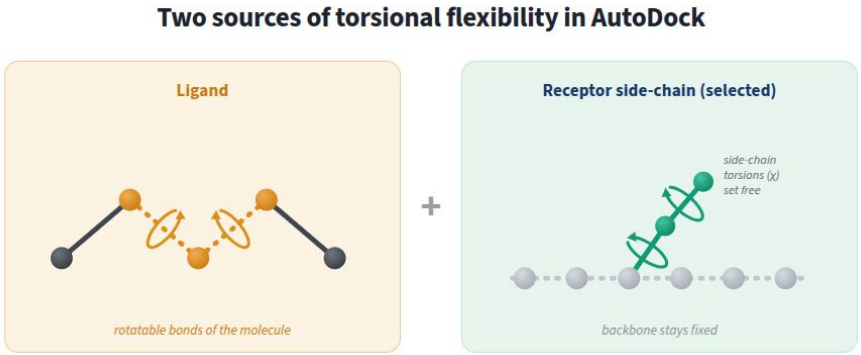
Adding Flexible residues - Autodock & GNINA

How it works

- Receptor is split into a rigid part + a few selected flexible side chains.
- Chosen residues keep their side-chain torsions free; the backbone stays fixed.

Trade-offs

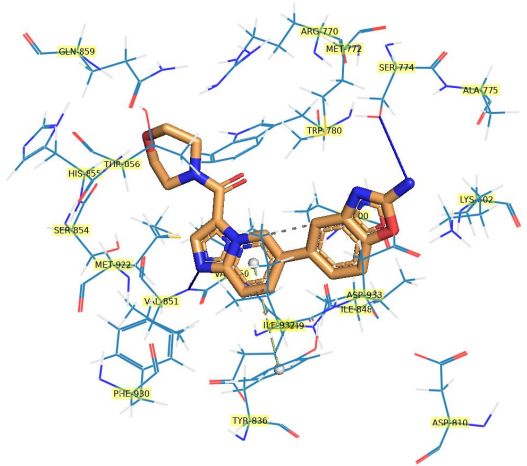
- Models local rearrangements in the pocket - more difficult comparison.
- Each residue adds rotatable degrees of freedom - search space & cost grow fast.



Adding Flexible residues - Autodock & GNINA

General Considerations

- Keep to a handful of pocket residues to limit noise and runtime.
- Consider increasing the exhaustiveness of the search if possible
- Explore the pocket: Choose residues that may be blocking the site



Flexible residues

Add flexible residues: Yes No

Residue chain: {"model": 0, "chain": "A", "residues": 1021}

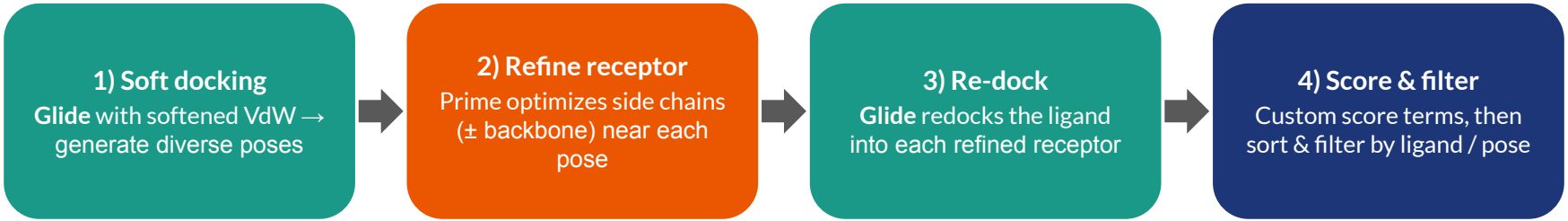
Flexible residues: {"index": "859-859", "residues": "Q"}

Add defined residues

List of flexible residues:
A:MET922
A:GLN859

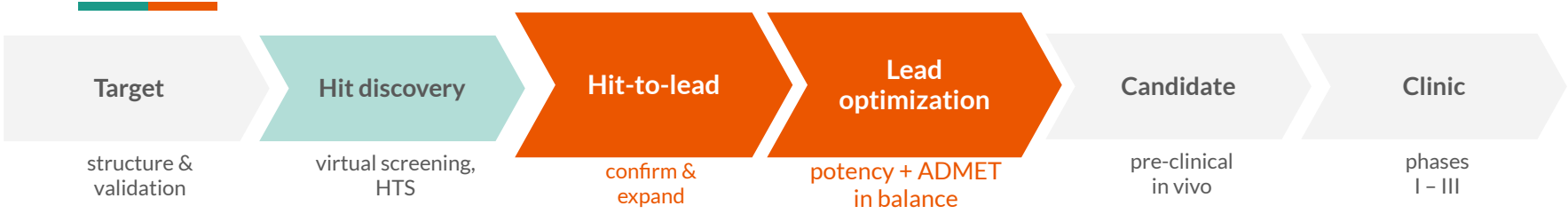
Induced Fit Docking - IFD Schrödinger

Schrödinger protocol that couples docking with receptor refinement to model induced fit — both side chains and local backbone.



- Protocols trade speed vs accuracy - best for a focused pose set.
- Commercial (Schrödinger suite).

2) Lead optimization



HIT

- Binds the target (sub-μM)
- Often poor ADME / selectivity

→

LEAD

- μM affinity, clear SAR
- Acceptable drug-likeness
- Room left to optimise

→

CANDIDATE

- nM potency and selectivity
- Good PK, safe in vivo
- Synthesizable at scale

Lead optimization turns a promising hit into a drug candidate through iterative chemical modifications, improving many properties at the same time.

Lead optimization in Scipion-Chem

Protocols that performs promising chemical changes on ligands in order to:

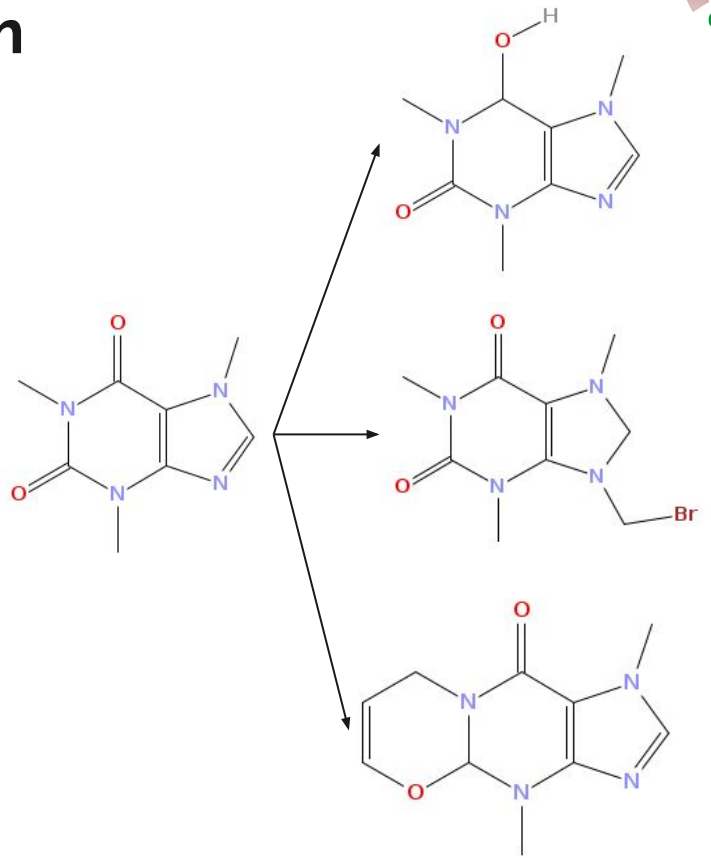
- Improve **affinity**
- Improve **ADME** or other **physicochemical features**
- Facilitate chemical **synthesis**

AutoGrow (DurrantLab)

DeepFrag (DurrantLab)

REINVENT4

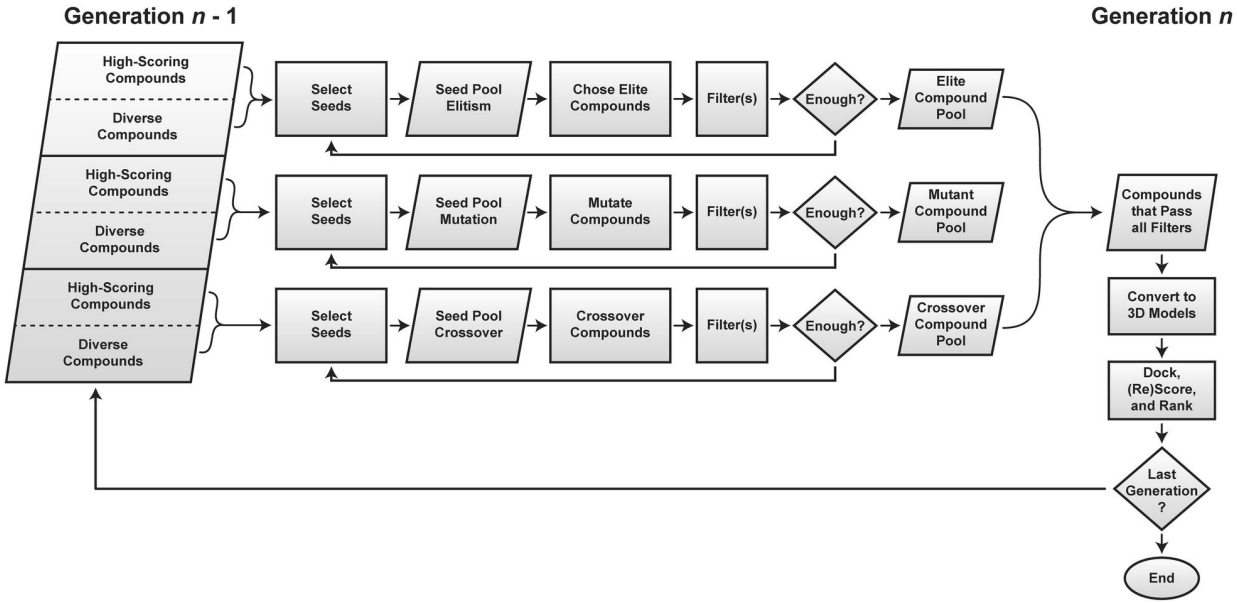
Flowr.root



AutoGrow4

Genetic algorithm (GA) for de novo design and lead optimization

1. **Seed population** - start from seed molecules (Hits) or fragments
2. **Dock & score** - predict binding affinity with docking
3. **3 operations: Select elites, mutate & crossover** - keep top-scoring (and diverse) parents
4. **Filter** - add drug-like filters
5. **Dock & score** - predict binding affinity with docking
6. **Select new seed population** - repeat from step 3



Spiegel, J.O., Durrant, J.D., 2020, J Cheminform

AutoGrow4

Transformations based on 3 approaches

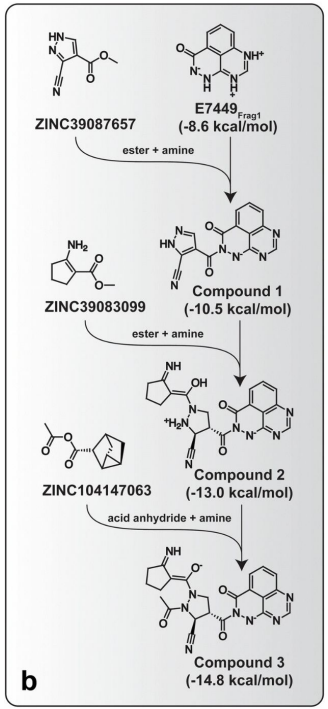
1. **Elitism:** Choose the fittest
2. **Mutate**
3. **Crossover**

Docking + Scoring using AutoDock Vina + Diversity Score

User define the number of iterations

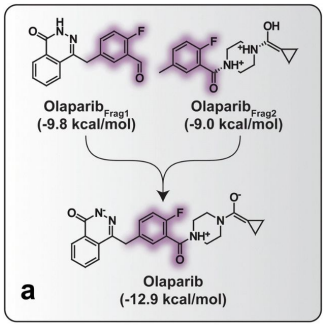
Mutation

+ fragment → grown molecule



Crossover

parent A + parent B → hybrid



Spiegel, J.O., Durrant, J.D., 2020, J Cheminform

DeepFrag

Machine learning model that generate **variations** that optimize the **affinity** of **docked ligands**

Parent + site

ligand with a fragment removed at a chosen connection point

Voxelize

3D grid of atom-density channels around that point

3D CNN

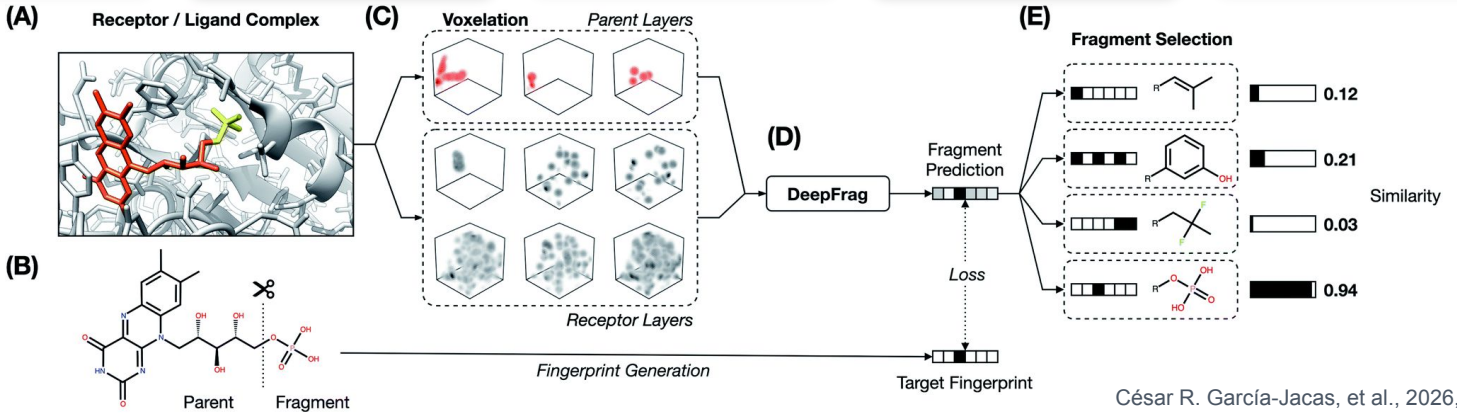
predicts a fingerprint of the ideal fragment

Retrieve

match fingerprint against a fragment library

Ranked output

best fragments to add, ranked by similarity



César R. García-Jacas, et al., 2026, Digital Discovery

DeepFrag

Two options:

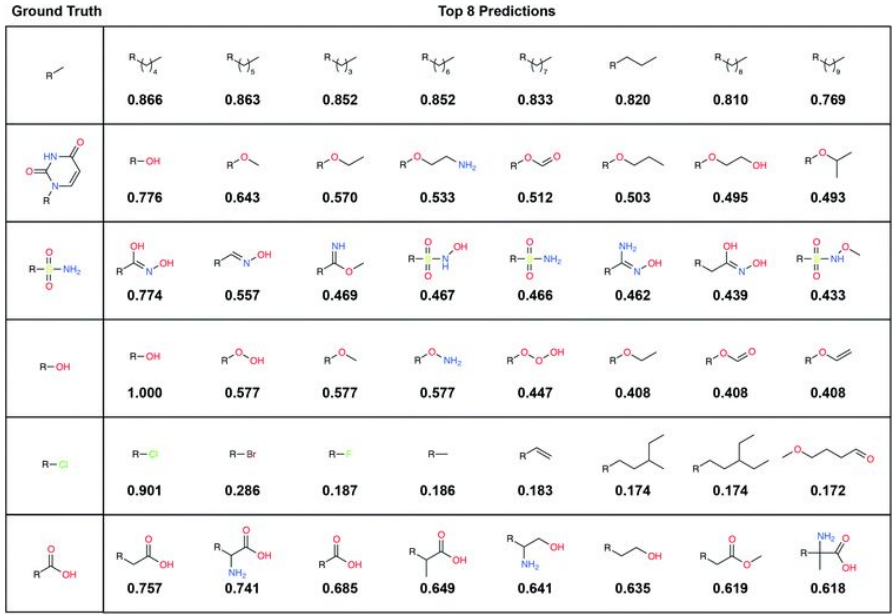
1. Substitute a fragment
2. Grow the molecule - add fragments

Pros:

- Super fast
- User decide where to grow or which fragment to substitute

Cons:

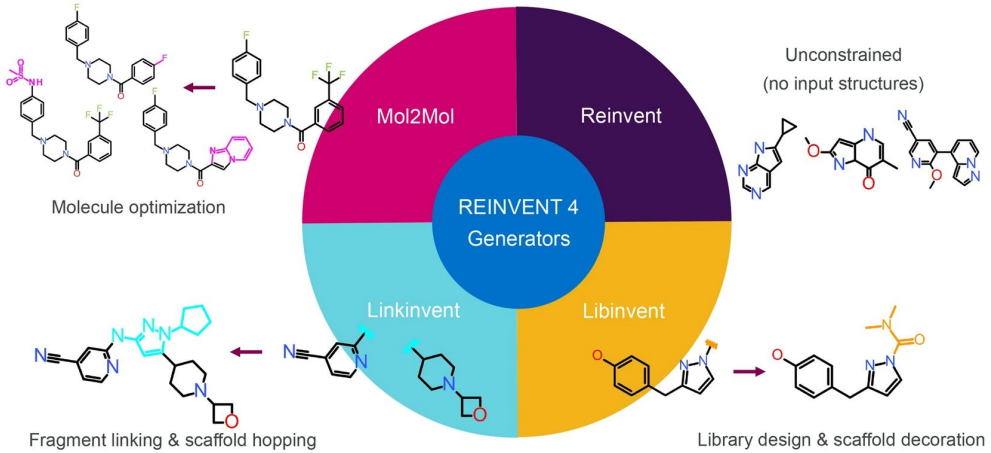
- Does not give any energy output
- Needs to perform docking afterwards to get the pose



César R. García-Jacas, et al., 2026, Digital Discovery

REINVENT4

Generative AI model that designs new molecules and improves existing ones, steered by reinforcement learning



Loeffler et al., 2024, J Cheminform

Optimization loop

1. Generate
A pretrained prior proposes new molecules as SMILES

→

2. Score
A user-defined scoring function rates them: docking score, drug-likeness, similarity…

→

3. Learn
Reinforcement learning shifts the generator toward high-scoring chemistry

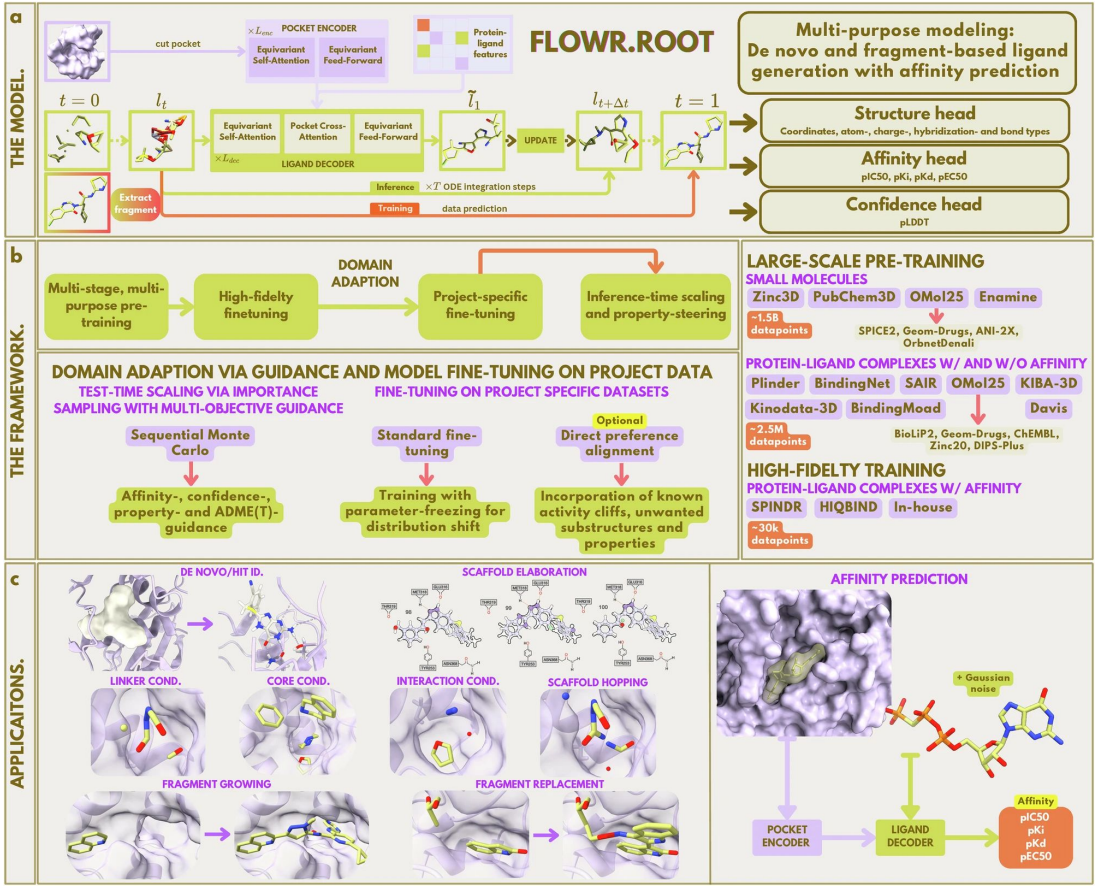
↺

FLOWR.ROOT

Flow-matching foundation model.

A lot of applications:

- **De novo**: invent chemical matter in the pocket
- **Interaction-conditional**: hold named contacts
- **Scaffold hopping**: swap the core, keep decorations
- **Scaffold elaboration**: keep core, regrow R-groups
- **Core & fragment growing**: build outward, aimed
- **Substructure replacement**: by atom index



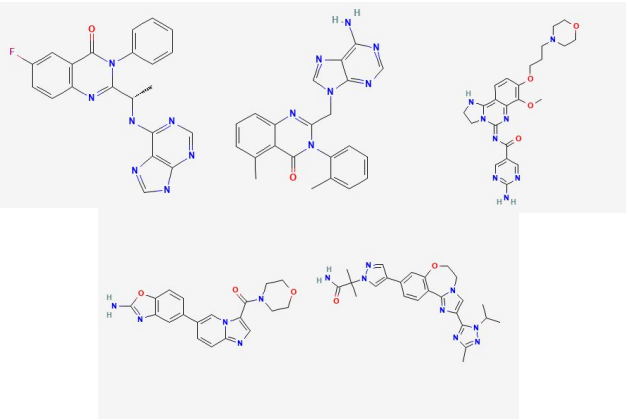
Cremer J., et al., 2026, Nat. Commun.

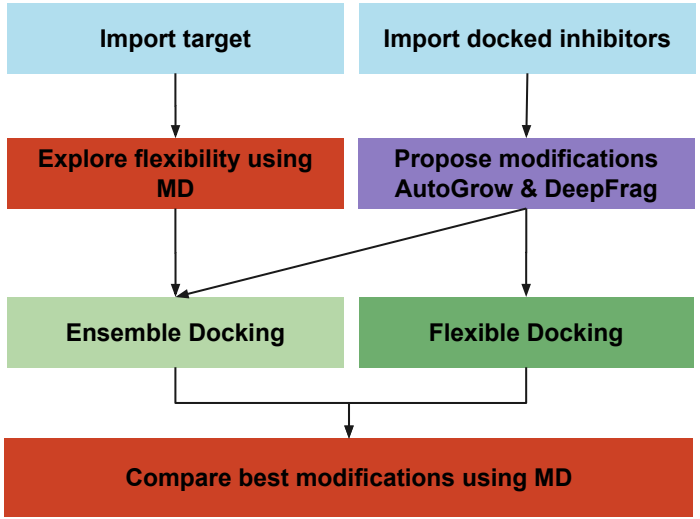
HANDS-ON: Lead optimization of PI3Kα inhibitors

Project: **FineTuningPI3Ka_course**

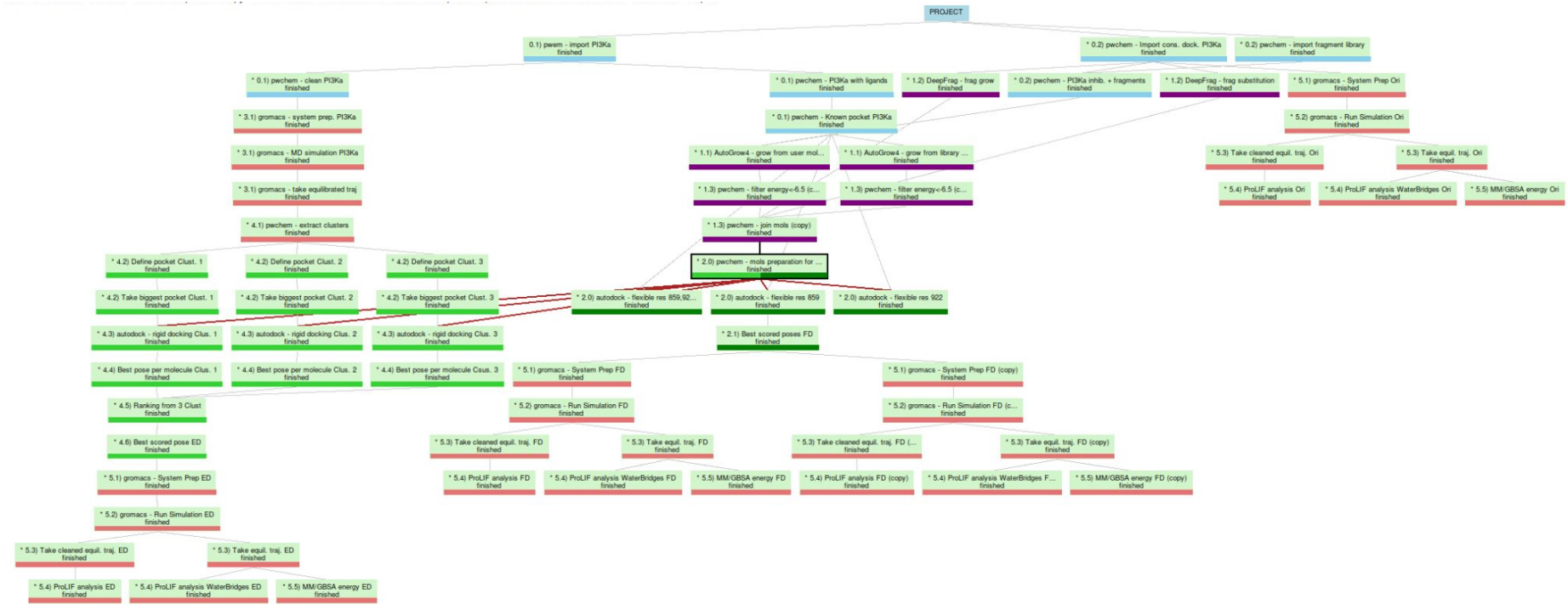
Objective: propose molecules from the PI3Kα inhibitors with increased affinity

1. Propose new molecules from the consensus docking outputs of the **Selective Docking** project
2. Flexible docking to PI3Kα





HANDS-ON: Lead optimization of PI3Kα inhibitors



Proposed exercises

1. Choose a different molecule from the inhibitors of PI3Kα (imported cons. dockings) to optimize
    a. Check the pocket residues to think of good atoms to grow from or replace
    b. Run DeepFrag to propose new modified molecules
    c. Explore the results and check if the molecule is growing in the atom(s) you want
2. Try flexible docking adding alternative flexible residues
3. Explore the results and study if the generated molecules give reasonable scores